PROJECT AND TEAM INFORMATION

Project Title:

Second-Hand College Item Exchange System

Team Name: *Team- DSCPP-III-2026-T352*	
Team member 1 (Team Lead) *SACHAN PARIDHI, student ID: 2510018933, paridhisachan27@gmail.com*	
Team member 2 *CHAUDHARY ROHAN, student ID: 2510380044, chaudharyyrohan01@gmail.com*	
Team member 3 *ARORA KANAV, student ID: 2510012486, kanavarora0808@gmail.com*	

PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

The Second-Hand College Item Exchange System is designed to address the accumulation of reusable items among college students, such as used books, calculators, lab equipment, stationery, and bags. The primary objective is to provide a structured, user-friendly platform where students can seamlessly list, browse, search, sort, buy, exchange, and donate used college items. By replacing informal and undocumented methods of exchanging goods, the project promotes economical reuse while practically demonstrating core concepts of Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOPs)

Updated Project Approach and Architecture

While the initial proposal outlined a command-line interface (CLI), the project approach has been successfully upgraded to feature an intuitive GUI-based application. The architecture now consists of a robust frontend graphical interface for user interaction and a powerful backend core engine handling authentication, item management, searching, sorting, and request processing. The system incorporates foundational DSA structures (Linked Lists, Queues, Stacks, Searching, and Sorting algorithms) and OOP principles. Persistent file-based storage using native file I/O methods is maintained for users, items, exchange requests, and history logs.

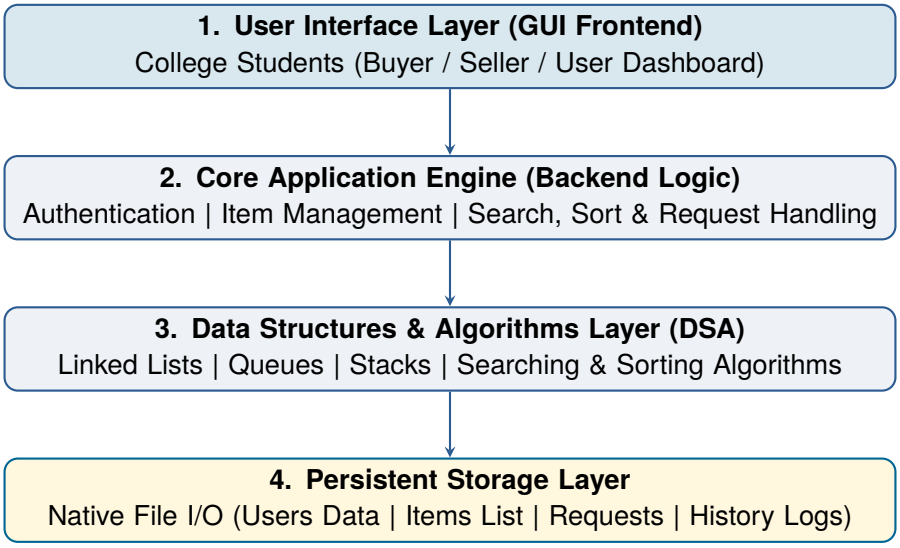


Figure 1: High-Level System Architecture Diagram

Tasks Completed

Task Completed	Team Member
1. Requirement analysis and system workflow finalization. 2. Implementation of core backend data structures (Linked Lists, Stacks, Queues) and logic. Development of the graphical user interface (GUI) replacing the initial CLI design. 3. Integration of file-based persistent storage for user data and item listings.	Paridhi Sachan Rohan Chaudhary Kanav Arora

Challenges/Roadblocks

Transitioning the application design from a command-line interface (CLI) to a modern GUI-based structure presented initial architectural challenges regarding event handling and seamless data binding between frontend views and backend C/C++ logic. Additionally, we had to deal with the problem of efficient file-based persistence for concurrent item requests, which involved a lot of debugging of custom search and sort routines. We solved these issues by breaking the code into distinct layers for UI and logic with event callbacks and extensive testing of file read/write not to corrupt the data..

Tasks Pending

Task Pending	Team Member (to complete the task)
1. Documentation finalization and preparation of the final project report. 2. Final polishing of GUI layouts and user feedback notifications. 3. Comprehensive testing and validation against edge cases.	Paridhi Sachan Rohan Chaudhary Kanav Arora

Project Outcome/Deliverables

Key results include the creation of desktop interface, which allows the students to sign-up/login, publish items, search and sort them, make buying/exchange/donation requests. Also, there is a file-based storage database, full documentation and validation test suite, which prove the correct implementation of OOP and DSA concepts.

Progress Overview

The progress is around 80 percent complete for now. The main backend data structures and file storage routines are fully developed and integrated with the frontend GUI (which is ahead of the CLI prototype's creation milestone). It should be noted that the current progress is consistent with the initial plan.

Codebase Information

Repository: Local/Version Control System repository for DSCPP Phase-II
Branch: `main` / `gui-development`
Important Commits:
- `init`: Initial setup of project architecture and data structures.
- `backend`: Implementation of file I/O storage and request-handling logic.
- `frontend-gui`: Transitioned layout from command-line interface to desktop GUI components.

Testing and Validation Status

Test Type	**Status (Pass/Fail/In progress)**	**Notes**
Unit Testing (DSA Algorithms)	In progress	Verifying search and sort algorithms with mock item datasets.

Deliverables Progress

1. Working GUI Application: Completed
2. Backend Database & File Storage: Completed
3. DSA/OOP Implementation: Completed
4. Testing, Validation, and Documentation: In Progress